

0:00–0:20

A paper from Nature Communications, published at the end of September 2026.

The authors train generative models on electron backscatter diffraction (EBSD) maps of 25 variants of one alloy, and search them for microstructures that are both stronger and more ductile.

I follow the paper's own story.

Outline

1	**The design problem**	Why ask a generative model for new microstructures?
2	**Groundwork**	Which methods does the paper build on?
3	**The method**	How are microstructures generated and steered toward properties?
4	**The results**	What do the authors show?

0:20–0:50

Four sections, each answering one question.

Section 2 introduces the machine-learning methods the paper combines, with their origins, for anyone who has not followed that literature.

Sections 3 and 4 present the paper as the authors tell it.



1:00–2:10

Each point is one alloy: Inconel 625, a nickel-based superalloy used for turbine discs, blades and combustors, processed in one of 25 ways.

The alloys differ only in processing: hot rolling and annealing at different temperatures and times.

Each was tensile-tested for its yield strength and its elongation to failure.

The stronger variants stretch less: a strength–ductility trade-off; making a metal stronger usually makes it brittle, and making it ductile makes it weak.

The dashed staircase is the front: no measured alloy beats a point on it in both properties.

The goal is a microstructure above and to the right of the front.

Such microstructures are usually found by experience and trial and error over processing routes; the paper asks a generative model to propose them.



2:10–3:10

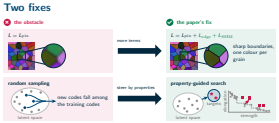
These are real patches of 50 by 50 micrometres from test alloys, cut from EBSD maps.

In an EBSD map each pixel's colour encodes the crystal orientation there: the three Euler angles as red, green and blue; the black lines are grain boundaries.

The second row is what the paper's generator returns when it is trained to reproduce its input with a pixel loss alone, the mean absolute difference of the pixel values (MAE); the mean squared difference (MSE) is the other classic choice, and both average over all pixels.

The boundaries blur, and colours mix inside grains, although a real grain has nearly one orientation.

Boundaries cover few pixels, so an average over all pixels barely notices them; yet boundaries and orientations are what set the mechanical properties.



3:10–4:10

The paper answers two obstacles with two changes: the obstacles on the left, its fixes on the right.

First, a pixel loss alone blurs grain boundaries, as the enlarged region shows; the paper adds two terms, one that responds to grain boundaries and one to the structure inside grains.

Second, codes drawn at random from the model's latent space fall among the training codes, so the new samples resemble the measured alloys.

So the authors search the latent space, steered by predicted strength and ductility, toward targets outside the training codes, as their Fig. 1c sketches; their five candidates lie beyond the measured front.

Section 2 explains the models and the search for anyone who has not met them.

Three ways to generate an image



4:20–5:20

Generative models differ in how they build an image.

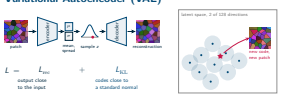
In one pass: a decoder turns a short code into the whole image at once; VAEs and GANs work this way.

Token by token: a network predicts each piece of the image from the pieces before it, as language models write text word by word; DALL-E 1 (Ramesh and colleagues, 2021) wrote 32 by 32 such tokens, which a discrete VAE then decoded into the image.

Such a model starts from scratch and works in a fixed order, so it cannot go back and refine what it has written.

Step by step from noise: a diffusion model starts from noise and refines the whole image at every step; Stable Diffusion and DALL-E 2 and 3 work this way, and diffusion leads image generation today.

The paper joins the first and the third: a VAE generates in one pass, and its output guides the diffusion steps.



5:20–6:40

An autoencoder is a network trained to copy its input through a bottleneck: an encoder compresses the image into a few numbers, the latent code, and a decoder rebuilds the image from them.

The variational autoencoder, from Kingma and Welling in 2013, makes the code probabilistic: the encoder gives a mean and a spread for each number, and the decoder sees a value drawn from that range.

A Kullback–Leibler (KL) term pulls every code towards a standard normal distribution, so the codes of different images overlap and the space has no gaps.

That makes a VAE generative: a code drawn from this space decodes to a plausible image, and moving through the space changes the image smoothly.

Trained with a pixel loss, a VAE returns blurry images, the average of the plausible answers: the failure of slide 3.

In this paper each patch gets a code of 128 numbers.



6:40–7:40

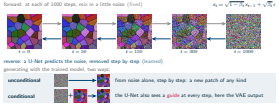
A trained generative model can do three different things, and each needs a different check.

Reconstruct: encode a training patch and decode it again; the result can be compared with its own input, pixel by pixel or by its statistics.

Generate: decode a new code from among the training codes; the result has no original, so it is compared with the statistics of held-out data and checked for near-copies of training patches.

Extrapolate: decode a code away from all training codes; no example lies there, so only an experiment can tell whether the result is realistic.

The paper claims all three; the results show what each claim rests on.



7:40–8:50

A diffusion model learns to undo noise.

The forward process is fixed: each of 1000 steps scales the image down slightly and adds a little Gaussian noise ϵ, with β_t rising from 0.0001 to 0.02; after a few hundred steps only noise is left.

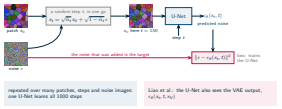
Each step depends only on the one before, a Markov chain; the paper writes it as a Gaussian, its Eq. 12.

The reverse process is learned: a network predicts the noise in an image at any step, and removing it step by step runs the process backwards.

The idea is from Sohl-Dickstein and colleagues, 2015; DDPM, from Ho, Jain and Abbeel in 2020, made the images sharp.

Once trained, it generates in two ways: from noise alone, any new patch; or with a guide image that the U-Net sees at every step, a patch that follows the guide.

The paper's guide is the VAE's blurry output, so the diffusion model only restores detail; the next slide shows the training.



8:50–10:00

A diffusion model is trained on one simple task: predict the noise in a noised image.

Take a training patch x_0, draw a noise image ϵ and a random step t, and mix them in one go; $\bar{\alpha}_t$, the product of all $1 - \beta_s$ up to step t, says how much of the patch is left.

The network, a U-Net (Ronneberger and colleagues, 2015), gets the noised patch and the step and returns a noise image of the same size.

The loss is the squared error between the added and the predicted noise; repeated over many patches, steps and noise images, one network learns every step.

Knowing the noise is enough to step back: each reverse step subtracts a share of the predicted noise and adds a little fresh noise.

The paper trains its diffusion model this way, with the VAE output x_V as a third input, and compares the noise with the loss of section 3.



10:00–11:20

With two goals that fight each other, such as strength and ductility, there is rarely one best answer, only a menu of trade-offs.

NSGA-II, from Deb and colleagues in 2002, searches for that menu, and its name says how.

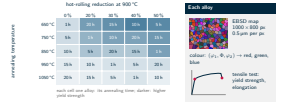
GA, a genetic algorithm (Holland, 1975): a population of candidates evolves over generations; each is scored on both goals, the best become parents, and children mix their parents' parts (crossover) with small random changes (mutation).

NS, non-dominated sorting: A beats B because it is better in both goals; nobody beats A, C or D in both, so they form front 1, the Pareto front, after Vilfredo Pareto, 1906; without front 1, the unbeaten rest form front 2, and the best are taken front by front.

II, crowding distance: when a front has more candidates than places, near-copies are dropped and candidates with room around them are kept, so the population spreads along the whole front.

Section 3 shows how the paper runs this search.

25 alloys, one map each



11:30–12:40

The 25 alloys come from an orthogonal design: five hot-rolling reductions, five annealing temperatures and five annealing times, combined so that every pair of levels occurs once.

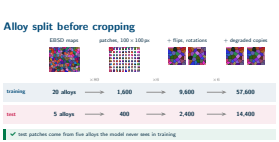
Each cell is one alloy; the number is its annealing time and the shade its yield strength.

More rolling and lower annealing temperatures give the stronger alloys.

Each alloy has one EBSD map of 1000 by 800 pixels at 0.5 µm per pixel, a field of 500 by 400 µm, with the three Euler angles of each pixel stored as red, green and blue.

Each alloy also has one yield strength and one elongation; how many tensile tests stand behind them is not stated.

So the dataset is 25 maps and 25 pairs of property values.



12:40–13:30

The split comes first: the 25 maps are divided at random into 20 training and 5 test alloys.

Only then is each map cut into 80 patches of 100 by 100 pixels, 50 by 50 µm.

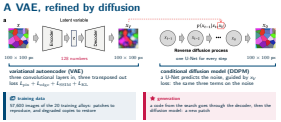
Flips and rotations make six versions of each patch.

Each version also gets five degraded copies (blur, distortion, black blocks, erosion and dilation), which the model learns to restore.

That gives 72,000 images: 57,600 for training and 14,400 for testing.

Because the split is by alloy, no test patch shares a map with a training patch: the same rule as splitting by specimen in segmentation.

The images multiply; the alloys stay 25.



13:30–14:20

This is the paper's own diagram of its generator, Fig. 1a.

The VAE compresses each patch of 100 by 100 pixels into a code z of 128 numbers and decodes it into x_V, which is smooth.

A conditional diffusion model then refines x_V: its U-Net predicts the noise step by step and sees x_V at every step, the pairing of DiffuseVAE; the red arrow marks that condition.

Both models are trained with the pixel, edge and SSIM terms of the next slide: the VAE on its output, the diffusion model on the noise it predicts.

Training uses the 57,600 images of the 20 training alloys, with Adam over 100 epochs.

The code z is what the search moves around later.



14:20–15:30

The loss is the sum of three terms, added without weights.

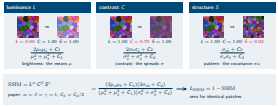
The pixel term is the mean absolute difference between original and reconstruction; boundaries cover few pixels, so it hardly sees them.

The edge term applies the Sobel filter, a three-by-three derivative filter from Sobel and Feldman, 1968, across and down both images and compares the responses: it is large where colours change, at grain boundaries.

The structure term is one minus the structural similarity index (SSIM) of Wang and colleagues, 2004, which compares the means, contrasts and correlation of the two images; the authors use it for the orientation inside and between grains.

The VAE adds its Kullback–Leibler term; the diffusion model applies the same three terms to the noise it predicts.

All three compare colours, that is, Euler angles read as intensities.



15:30–16:40

SSIM compares two patches in three ways, each through one statistic of the pixel values.

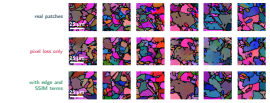
Luminance compares the means: a brighter copy of the patch scores $L = 0.89$, while C and S stay near 1.

Contrast compares the standard deviations: a flattened copy scores $C = 0.70$.

Structure compares the patterns through the covariance: shuffling the patch's tiles keeps its mean and spread but drops S to 0.02.

The product, with all exponents 1 and $C_3 = C_2/2$ as in the paper, gives its Eq. 8; the loss is one minus SSIM.

The brighter copy differs by 0.24 per pixel on average and keeps SSIM 0.89; the blurred patch of slide 4 differs by only 0.05 but drops to 0.84: SSIM sees lost structure that a pixel loss barely notices.



16:40–17:20

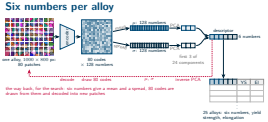
The same six test patches, with a third row: the model trained with all three terms.

The boundaries are sharp again, and each grain has nearly one colour.

The pixel loss itself also falls, from about 0.20 to about 0.10 according to the text; the plotted curve settles nearer 0.07.

Two more reconstruction terms also weaken the Kullback–Leibler term relative to the rest, which by itself can lower a VAE's reconstruction error; the paper does not separate the two effects.

The comparison with a standard VAE, a Wasserstein GAN and a standard DDPM is only in the Supplementary Information.



17:20–18:10

To link microstructures with properties, each alloy needs one descriptor.

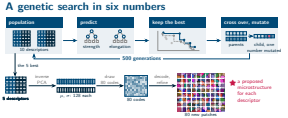
The 80 patches of its map are encoded, which gives 80 codes of 128 numbers.

Their mean and standard deviation over the patches describe the alloy: what its patches look like on average, and how much they vary.

Principal component analysis (PCA, Pearson 1901), fitted on the 25 alloys, keeps the first three of 24 components of each, so every alloy becomes six numbers beside its measured properties.

PCA can be inverted, the red path: six numbers map back to a mean and a spread, from which new codes are drawn and decoded.

The number of components was chosen by the fidelity of the generated patches (Supplementary Information).



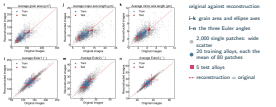
18:10–19:00

The loop of section 2, run on the six numbers.

Two gradient-boosted tree models, one per property, fitted on the 20 training alloys, predict yield strength and elongation from a descriptor; they are the objectives.

NSGA-II evolves a population of 10 descriptors over 500 generations; the paper's average hypervolume improvement, the area the candidates span above the weakest measured alloy, levels off after about 150 generations.

The five best final descriptors go back through the inverse PCA to a mean and a spread; 80 codes are drawn from them and decoded into patches, the proposed microstructures.



19:10–20:00

Six features are measured in each original patch and its reconstruction: grain area, the major and minor axes of the equivalent ellipse (panels i to k), and the three Euler angles (l to n).

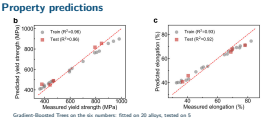
The grey points are the 2,000 single patches; their means scatter widely, more so for large grains, of which one patch holds only a few.

Averaged over the 80 patches of each alloy, the 20 training alloys in blue and the 5 test alloys in red lie on the line of equality.

So the model reproduces the grain statistics of alloys it never saw.

What is compared: reconstructions with their own inputs, the first task of slide 7.

Misorientation statistics and the baseline models are in the Supplementary Information only.



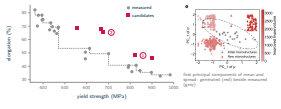
20:00–20:40

These are two gradient-boosted tree models (Friedman, 2001), one per property: sums of many small regression trees on an alloy's six numbers; the number of trees and their depth were set by a randomized search.

On the five test alloys, R^2 is 0.96 for yield strength and 0.92 for elongation; on the 20 training alloys 0.96 and 0.93.

R^2 is the share of the variance of the measured values that the predictions explain.

These two models are what the search optimizes.



20:40–21:40

After 500 generations the search returns five candidates, here with their predicted properties.

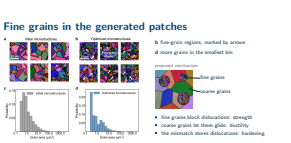
All five lie above the measured front: at equal strength, they are predicted to stretch more.

Their predicted values stay inside the ranges of the 25 measured alloys; the gain lies in their combination.

On the right, the first principal components of the mean and the spread: measured alloys in grey, generated microstructures as triangles, darker for a larger hypervolume improvement.

The best candidates lie far from every measured alloy; the authors call them out of distribution, the third task of slide 7.

Candidates 1 and 2 are the ones checked by simulation.



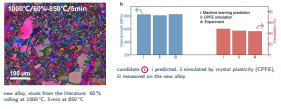
21:40–22:40

Left, patches of the initial alloys and patches generated for the candidates; the arrows mark what the authors call fine-grain regions (panel b).

Below, the grain areas of all 2,000 original patches and of 400 generated ones, on a logarithmic axis: the generated patches have more grains in the smallest bin, at or below 0.4 μm^2 (panel d), which the authors read as a bimodal grain size.

Their explanation: fine grains block dislocations and raise the strength, coarse grains leave room for them to glide and keep the ductility, and the mismatch between the two stores geometrically necessary dislocations (GNDs), which raise the strain hardening.

The mechanism is established for heterogeneous microstructures; whether these generated patches show it remains to be checked.



22:40–23:40

Two checks follow.

First, crystal plasticity finite element (CPFE) simulations of two generated patches per candidate, with a slip model calibrated on three measured stress–strain curves; the Euler angles come from the pixel colours, the dislocation density from a learned mapping.

Second, one experiment: an alloy processed by a route the authors chose from the literature to get a bimodal grain size, 60 percent hot rolling at 1000 °C and 5 minutes at 850 °C.

Its EBSD map, left, shows coarse grains among many fine ones.

Right, candidate 1 as predicted, simulated and measured: yield strength about 820, 805 and 825 MPa, elongation about 48, 44.5 and 43.5 percent, read from the bars.

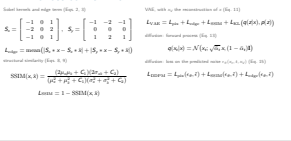
The paper concludes that the design is validated.

23:40–24:00

Thank you.

Questions to open the discussion: would edge and SSIM terms help our segmentation of grain boundaries; what minimum grain size do we use in EBSD analysis, and why; how many independent specimens would we want before trusting a property model; and what would convince us that a generated microstructure is new and can be made.

Backup slides follow: the loss terms in full, the 25 processing routes and the references.



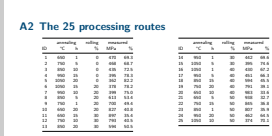
If asked

The pixel, edge and SSIM terms are added without weights, in the VAE and in the diffusion model.

In the diffusion model they compare the predicted with the true noise, so their link to grain boundaries is indirect.

Eq. 14 as printed gives the predicted noise as the mean of the reverse step; standard DDPM sampling derives that mean from the predicted noise.

SSIM uses its three factors (luminance, contrast, structure) with equal exponents of one, which gives Eq. 8; C_1 and C_2 are small constants.



If asked

The table of the code repository: each variant's processing route, hot rolling at 900 °C and then annealing, with its measured yield strength and elongation.

Every pair of levels occurs once: an orthogonal design.

Alloy 10 is the measured alloy closest in strength to candidate 1.

The paper's 20/5 split is random and not listed; from Fig. 4b, the test alloys lie near 375, 445, 470, 795 and 845 MPa.